

Group 14 Evaluation Report: SmartSecurePerimeter (SSP)

Detailed Technical Assessment against P1 & P2 Criteria | IoT Design Project 2026/2027

MANDATORY SUBMISSION RULES & FORMATTING REQUIREMENTS

- **CRITICAL WARNING — FORMAT & LENGTH NON-COMPLIANCE:** Check the uploaded deliveries from previous years and the official template (Template_for_Groups_Delivery_Correction_26-27_v0.pdf) immediately. Your 45-page document is completely wrong in format and structure. Deliverable P2 **MUST** follow the standard 2-column IEEE transaction template and be concise (target 4–8 pages). Move exhaustive requirement lists to 1-column annexes.
- **File Naming & Format:** All future deliveries must strictly follow the format Group14_SSP_PY.pdf (PDF format for submission) and Group14_SSP_PY.docx (for editable source). Current file 'D2_Functional_Performance_Specs_Group_14.pdf' uses an invalid naming scheme.
- **Incremental Yellow Highlight:** In subsequent deliveries (P4–P6), teams must explicitly mark all modified or newly added text, tables, and figures with a yellow background highlight. Do not do it for P3 as it would be all yellow!
- **Platform & Protocol Neutrality:** Do NOT lock in specific commercial ICs or protocols during P1/P2. Specify platform classes (e.g., ULP MCU, short-range wireless radio, wide-area gateway class) and quantitative performance specifications instead.

Project Identifier	SmartSecurePerimeter (SSP) — Perimeter Security & Personal Protection
Current File in Repository	D2_Functional_Performance_Specs_Group_14.pdf (⚠ Needs rename to Group14_SSP_P2.pdf)
Team Members & Degrees	Louis-Marie Loe, Cheong Travis (Group 14 UAB IoT Design Project 2026/2027)
Assigned Challenge	Original Project Idea (Smart Perimeter Security & Sensitive Zone / Personal Safety Monitoring)
Evaluation Focus	P1 (Concept & Market) and P2 (Functional & Performance Specifications)

1. Document Template, Formatting & Operational Storyboard

Good Quality / Strengths: Ambitious and thorough formulation of a multi-tier perimeter security and personal safety monitoring platform (protecting sensitive zones, schools, and individuals with restraining orders). The proposal demonstrates an impressive depth of engineering categorization (dividing requirements into F-Dxx, F-Exx, F-Cxx, F-CLxx, F-Uxx, F-AIxx and performance categories).

CRITICAL FORMATTING DEFECT — Excessive Document Length & Wrong Template: Please inspect uploaded deliveries from previous years immediately. Submitting a **45-page monographic**

document is completely non-compliant for Deliverable P2. The course guidelines explicitly mandate the official **IEEE two-column transaction template**. You must reformat the document into a concise core report (target 4–8 pages) that covers the introduction, market benchmark, summary specifications, and system diagrams. Move exhaustive list tables (such as the 100+ individual requirement lines) to **single-column annexes** at the end of the document.

MISSING REQUIREMENT — Operational Storyboard: Although Section 2 describes various operational use-case scenarios (Normal Monitoring, Geofence Approach, High-Risk Event, Connectivity Loss), the report lacks a clear, step-by-step **Operational Storyboard** illustrating end-to-end interactions across user personas (Protected Person / Victim, Security Operator, System Administrator). The team must include an explicit storyboard matrix mapping the user/system journey across distinct phases such as:

Phase	System & Sensor Action	User & Operator Interaction
1. Device Pairing & Geofence Setup	Wearable node pairs with mobile/edge gateway via low-power wireless; configures geofence boundaries and security credentials.	Protected person equips lightweight wearable node; security operator assigns monitoring role via Cloud Portal.
2. Normal Low-Power Telemetry	Inertial (IMU) and position sensors operate at low duty cycle (0.1–1 Hz); local pre-processing buffers status.	Protected person moves normally; dashboard displays green 'Normal State' with battery and connectivity health.
3. Perimeter Approach / High-Risk Event	Pre-processing detects rapid approach to restricted geofence or high acceleration; triggers Edge AI evaluation (<200 ms).	System increases sampling rate (50–100 Hz); mobile app vibrates discrete haptic alert to notify user.
4. Priority Alert & Cloud Dispatch	Edge confirms perimeter breach or tamper event; bypasses normal queue to transmit priority payload (<5s latency).	Security operator UI displays urgent map alert with location, risk score, and automated response workflow.
5. Offline Continuity & Re-Sync	In event of cellular disruption, Edge buffers up to 24h of telemetry and local risk decisions without data loss.	Local monitoring persists seamlessly; automatic back-pressure re-sync occurs once connectivity is restored.

2. Multi-Tier Architecture & AI Strategy Breakdown

Good Quality / Strengths: Excellent multi-tier architectural vision establishing clear boundaries across Device, Edge, Cloud, UI, and Security layers. The inclusion of offline buffering requirements (P-D12 ≥24h local buffer) and privacy-aware data minimization principles is commendable.

CRITICAL MISSING REQUIREMENT — Explicit AI & Edge AI Strategy Breakdown: Section 4.17 briefly mentions an AI accuracy target (P-AI01 ≥90%), but lacks a structured, multi-tier Artificial Intelligence architecture. In compliance with the 2026/2027 IoT curriculum expectations, the report must explicitly define the distribution of intelligence across tiers:

- Edge AI / On-Device Tasks (Wearable Node & Mobile Gateway):** Local real-time IMU motion pattern classification (detecting falls, sudden running, or physical struggle), local geofence trajectory

prediction, tamper detection, and contextual risk scoring (<200 ms latency) without requiring continuous raw sensor transmission to the cloud.

- **Cloud AI Tasks (Server Backend):** Multi-device spatial risk heatmaps, long-term movement pattern analytics, offender proximity correlation across multiple security feeds, automated alert dispatch, and cloud model retraining pipelines.
- **Model Optimization & Resource Constraints:** Explicit targets for model compression and quantization (e.g., INT8 quantization, TensorFlow Lite for Microcontrollers) targeting execution footprints under 50–100 KB RAM to preserve wearable battery life.

3. Market Benchmark & Economic Specification Correction (RRP vs. BOM & TCO)

Good Quality / Strengths: Section 3 provides a solid qualitative analysis of existing commercial trackers (T-Mobile SyncUP TRACKER, AngelSense, Jiobit), correctly identifying SSP's differentiation in multi-tier distributed processing, adaptive monitoring, and privacy-aware data minimization.

CRITICAL ECONOMIC CORRECTION — Target Retail Price (RRP) vs. Hardware BOM & Cloud Costs: The document currently lacks explicit cost metrics in Section 4. **As established across all IoT course deliveries, the economic performance specification in P2 MUST be the Target Retail Selling Price (RRP)** to the end user (e.g., Target RRP 80 € – 120 € per wearable unit), compared directly against retail competitor pricing. Raw hardware BOM manufacturing costs, cloud server hosting fees, and maintenance expenses must NOT be confused with the target retail price—they belong to the **Total Cost of Ownership (TCO)** analysis in Deliverable P6.

REQUIRED — Comparative Market Analysis Matrix (Table II): To establish a rigorous commercial baseline, the team must include a structured comparison matrix in Section 3 evaluating SSP against representative market solutions:

Solution / Product	Primary Mechanism	Key Features & Telemetry	Estimated Retail Cost	SSP Advantage / Gap
AngelSense GPS Tracker (aranetsense.com)	Cellular + GPS / Wi-Fi positioning + 2-way audio	Continuous tracking, geofencing, emergency voice listen, mobile app	\$99 – \$150 unit + \$20–\$45/mo sub	High recurring cost; continuous raw streaming drains battery; no Edge AI.
Jiobit Smart Tag (jiobit.com)	Multi-radio positioning (GPS, Bluetooth, Wi-Fi, Cellular)	Trusted places geofencing, proximity alerts, small form factor	\$130 – \$150 unit + \$8–\$14/mo sub	Good tracking; lacks local threat/motion classification and offline Edge AI.
T-Mobile SyncUP (t-mobile.com)	LTE-M cellular + GPS location tracking	Basic geofencing, light sensor, IP67 enclosure, mobile app	\$48 – \$96 unit + \$5/mo sub	Inexpensive; basic asset tracker without perimeter risk intelligence.
Proposed SSP System (Group 14 Solution)	Wearable IMU/Position + Edge AI + Cloud Security	Adaptive sampling, Edge AI risk inference, 24h offline buffer, privacy-first	Target BOM <35 € Target RRP 80–120 € Whole TCO <150 €	Combines adaptive low-power sensing, local Edge AI, and resilient offline security.

4. Quantitative Performance Specifications, Energy Budget & Environmental Specs

Good Quality / Strengths: Establishes solid quantitative latency targets: device acquisition latency <20 ms, local event decision <500 ms, device-to-edge range ≥ 10 m, end-to-end normal alert latency ≤ 10 s, and high-priority alert latency ≤ 5 s.

CRITICAL CORRECTION — Energy Budget Definition (Time Duty Cycles vs. Consumption):

Section 4.10 attempts to specify detailed power levels (1 mW deep sleep, 50 mW active, 150 mW sensing, 500 mW TX burst). **As established in course guidelines, teams must NOT detail explicit power (mW) or current (mA) figures in Deliverable P2**, because exact component power draws cannot be known until specific ICs are evaluated and selected in Deliverable P4. **Instead, the P2 Energy Budget must strictly specify:**

- **1. Operating Duty Cycles in Time:** Expected percentage of time spent in each operating state (e.g., Deep Sleep >95% of time, Sensing/Processing <4%, Wireless TX/RX burst <1%).
- **2. Target Battery Autonomy:** The required operational lifespan between charges (e.g., minimum 7 days continuous operation, extended objective 14 days), which will subsequently dictate battery capacity selection in Deliverable D4.

MISSING REQUIREMENT — Environmental, Chemical & Physical Durability Specifications: While Section 4.11 specifies maximum size (<100 cm³), weight (<100 g), drop resistance (1 m), and IP65 protection, the report must expand on crucial physical and chemical constraints for wearable safety devices:

- **Skin Biocompatibility & Hypoallergenic Standards:** The wearable enclosure and mounting clip must comply with ISO 10993 biocompatibility standards to prevent skin irritation during prolonged 24/7 contact.
- **Chemical & Sweat Resistance:** Enclosure materials (e.g., medical-grade ABS/PC or TPU) must withstand human perspiration, rain, and daily surface disinfection wipes (isopropyl alcohol) without material degradation.
- **Vandal & Tamper Resistance:** Physical enclosure must incorporate mechanical tamper-detection switches and impact resistance ratings (IK07/IK08) to withstand physical altercations or intentional destruction attempts.

ACTION PLAN FOR DELIVERABLE P3 / REVISED P2

1. Reformat Entire Document: Re-structure the 45-page document into a concise two-column IEEE format (4–8 pages). Move exhaustive requirement lists to single-column annexes.
2. Add Operational Storyboard: Insert an explicit step-by-step matrix mapping user/system interactions for all target personas across all operational phases.
3. Detail Edge & Cloud AI Strategy: Explicitly break down AI tasks between local Edge AI (motion pattern classification, risk evaluation <200ms) and Cloud AI (spatial heatmaps, pattern analytics).
4. Fix Economic & Energy Specifications: Replace raw BOM/cloud figures with Target Retail Selling Price (RRP: 80–120 €) and redefine the energy budget in terms of time duty cycles and target battery autonomy (7–14 days).
5. Correct File Naming & Highlights: Rename file to Group14_SSP_P2.pdf / Group14_SSP_P2.docx and apply yellow background highlights to all incremental changes in future deliverables.